

Group 2 - BME 212

Impact of Chronic Acidic Exposure on Bone Mechanical Strength: A Femoral Failure Analysis

Matt Chacon, Isaiah Coombs, Matthew Kuan,
Andrew Muzaka, Winston Ng

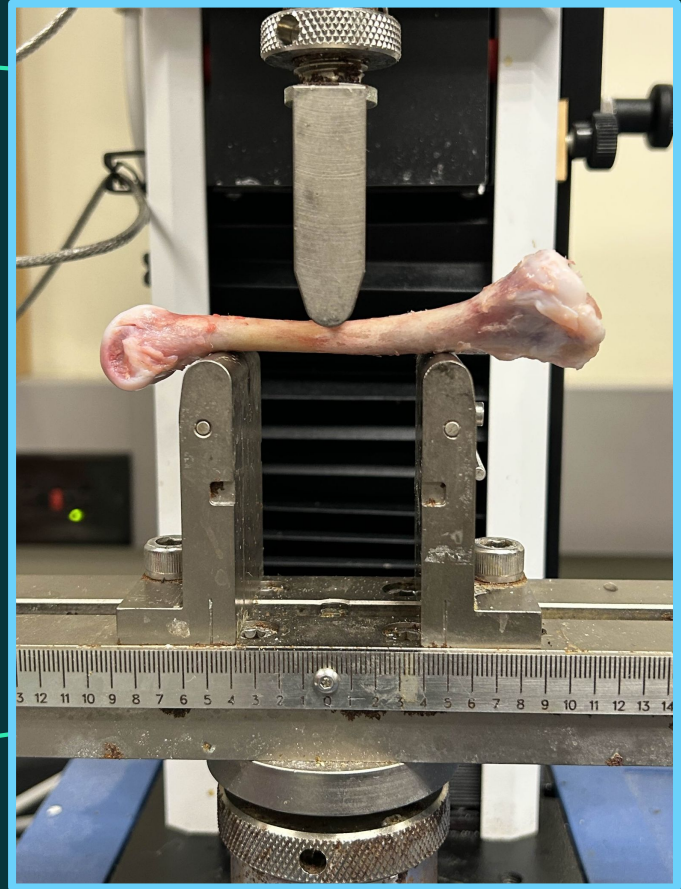
What do we know?

- When the human body experiences **fluctuations in body pH**, one of the first systems used as a **buffer** is the **skeletal system**.
- **Diseases can affect pH**, but so can **diet, genetics and lifestyle**
 - Are bone bending properties **significantly compromised** by **acidic environments**?

Our Goal:

- Gather **three-point bending** properties of **bones exposed to acid**
- Compare experimental properties to **control bones** (water)
- Determine a relationship between **bone quality** and **environmental acidity**

We hypothesize that chronic exposure to an acidic environment (≤ 3 pH) will reduce the femoral force of failure a bone can withstand by a minimum of 15% compared to the control group.



How was this performed?

Preparation

All muscle and skin were removed from **20 chicken legs** to isolate the femurs.

Matched femurs into pairs based on similar bone length (assumed left/right from the same chicken).

Assigned one femur in each pair to the acidic solution (**distilled white vinegar, pH 2.71**) and the other to the control solution (**tap water, pH 7.97**).

Experiment

Submerged all assigned bones in their respective plastic containers (**acidic** or **control** solution)

Kept bones fully submerged for **20 hours**.

Removed bones and placed them on paper-towels, allowing bones to dry for 30 minutes.

Conducted a three-point bending test on each femur using a 50 mm/min loading rate

Data Handling

Recorded force–displacement measurements during each bending test.

Uploaded all collected data to the designated Google Drive folder.

Created force vs. displacement graphs to find force-to-failure

Used paired data (treatment vs. control within each matched bone pair) for statistical analysis with a **paired t-test** ($\alpha = 0.05$).

Statistical Analysis

Test Used:

Two-tailed Matched Pairs T-Test

- Provides more significant results, and cleaner comparison
- Makes sense in context of before and after acid treatment
- Bones of closest size are paired

Power Analysis Table

	N ^b	Actual Power ^c	Power	Test Assumptions		Sig.
				Std. Dev. ^d	Effect Size	
Test for Mean Difference ^a	10	.803	.8	.15	1.000	.05

a. Two-sided test.

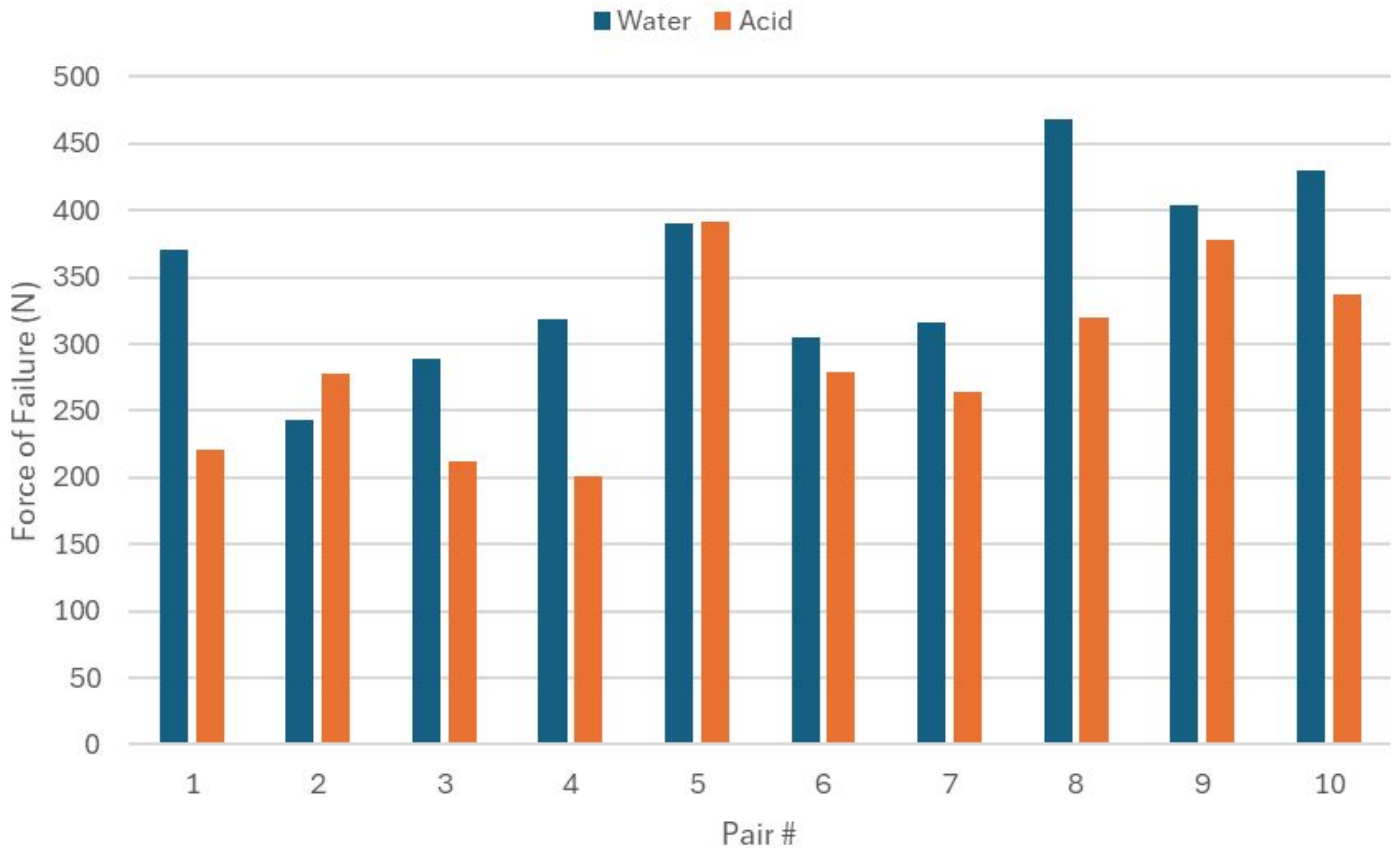
b. Number of pairs.

c. Based on noncentral t-distribution.

d. Standard deviation of the mean difference.

Sample size analysis: **10 matched pairs (20 bones)**

- Estimated mean difference of 15% based on literature finding difference of 10-20% between murine bones treated with and without acid [7]
- Estimated sample variance of 15%, conservative estimate based on study involving bone bending of over 2000+ bones (estimated 24-46% variability) [20]



What was found?

$$p = \underline{0.0092} \quad \alpha = 0.05$$

Results are Significant

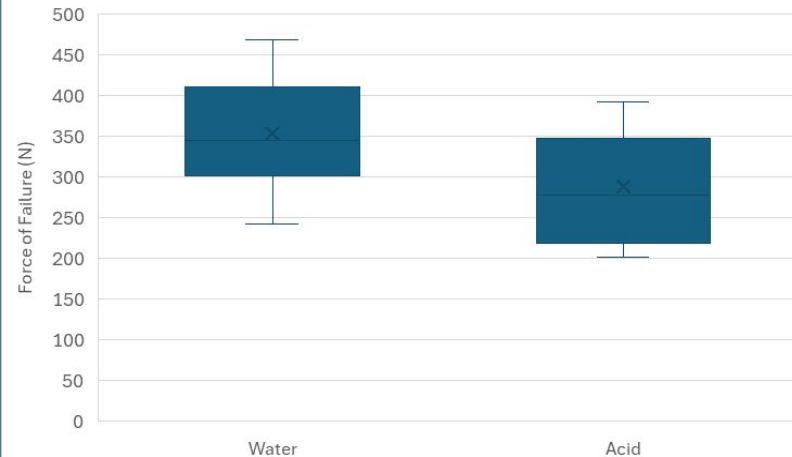
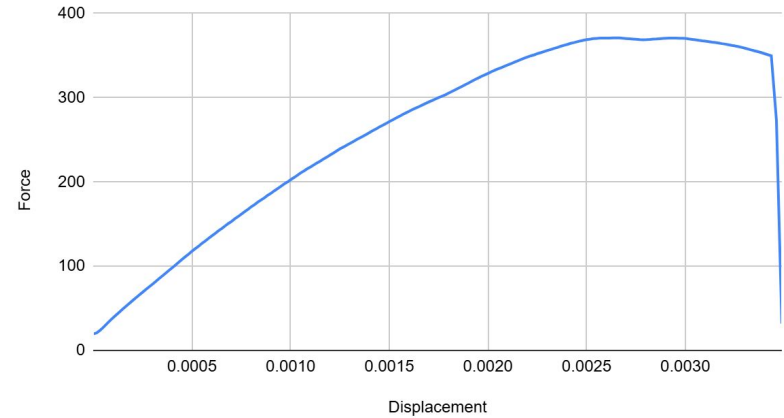
- Pairs 2 and 5 showed instances of acid group strength being **stronger** than water group strength
- Acid had lower mean, median and quartiles for force of failure than water group
- Both groups have **approximately normal** force of failure distributions (using *Shapiro-Wilk*)
 - $p > 0.8$

Analysis of Pair Differences

Mean paired difference: 65.6 N

SD paired difference: 62.8 N

Force vs. Displacement



How is this significant?

- We are able to reject the null hypothesis, so our hypothesis is supported
 - The p-value (0.0092) indicates statistical significance, suggesting that chronic acidic environments can measurably weaken bone strength.
- Acidic environments can lead to dissolution of bone minerals, reducing ultimate strength and increasing fracture risk.
- Dietary differences that influence systemic pH may contribute to long-term skeletal health outcomes.



Limitations

- **Experimental setup** does not fully replicate how acidosis affects bone tissue in the body
 - Vinegar contains organic compounds that may alter results
 - Bones stored at refrigerated temperature, not true body temperature
 - Delay between drying samples and mechanical testing may affect outcomes
- **Natural variation** in bone size, density, and porosity introduces variability
- **Limited resources** and small sample size due to budget constraints



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Thank you!
Q&A